

# Effective Heat Inactivation of *Mycobacterium avium* subsp. *paratuberculosis* in Raw Milk Contaminated with Naturally Infected Feces



The effectiveness of high-temperature, short holding time (HTST) pasteurization and homogenization with respect to inactivation of *Mycobacterium avium* subsp. *paratuberculosis* was evaluated quantitatively. This allowed a detailed determination of inactivation kinetics. High concentrations of feces from cows with clinical symptoms of Johne's disease were used to contaminate raw milk in order to realistically mimic possible incidents most closely. Final *M. avium* subsp. *paratuberculosis* concentrations varying from  $10^2$  to  $3.5 \times 10^5$  cells per ml raw milk were used. Heat treatments including industrial HTST were simulated on a pilot scale with 22 different time-temperature combinations, including 60 to 90°C at holding (mean residence) times of 6 to 15 s. Following 72°C and a holding time of 6 s, 70°C for 10 and 15 s, or under more stringent conditions, no viable *M. avium* subsp. *paratuberculosis* cells were recovered, resulting in >4.2- to >7.1-fold reductions, depending on the original inoculum concentrations. Inactivation kinetic modeling of 69 quantitative data points yielded an  $E_a$  of 305,635 J/mol and an  $\ln k_0$  of 107.2, corresponding to a D value of 1.2 s at 72°C and a Z value of 7.7°C. Homogenization did not significantly affect the inactivation. The conclusion can be drawn that HTST pasteurization conditions equal to 15 s at  $\geq 72^\circ\text{C}$  result in a more-than-sevenfold reduction of *M. avium* subsp. *paratuberculosis*.